MIL-V-43947A
2 March 1978
SUPERSEDING
MIL-V-43947(GL)
2 November 1976
(See 6.8)

MILITARY SPECIFICATION

VEAL CUTLETS, BONELESS, BREADED, FROZEN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers frozen, boneless, breaded veal cutlets for use by the Armed Forces in kitchens where facilities for frozen storage are available.

1.2 Classification. The product shall be texturized (flake-cut) breaded, boneless cutlets prepared from fresh dressed veal or calf of USDA Standard grade or better.

1.2.1 Shapes and weights.

I	Oval shape	- 5-1/2 ounces (156 g) (see figure 1)
II	Square shape	- 3-1/2 ounces (99 g) (see figure 2)
III	Other	- As specified (see 6.1)

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8905

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SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-1163	-	Box, Corrugated Fiberboard, High Compression Strength, Weather-Resistant, Wax-Resin Impregnated

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-668	-	Sanitary Standards for Food Plants

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

US DEPARTMENT OF HEALTH, EDUCATION AND WELFARE

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

US DEPARTMENT OF AGRICULTURE

Meat and Poultry Inspection Regulations

Official United States Standards for Grades of Veal and Calf Carcasses

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, Post Office Box 540, Benjamin Franklin Station, Washington, DC 22040.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Dept., 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

E 380 Standard for Metric Practice

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. This specification contains provisions for first article test and approval (see 4.1.2, 6.3 and 6.4).

3.2 Material.

3.2.1 Veal or calf. Bone-in veal or calf shall comply with the following requirements:

(1) Shall be prepared from chilled, fresh-dressed veal or calf carcasses, sides, quarters or wholesale cuts of USDA Standard grade or better. Veal or calf shall be graded in accordance with Official US Standards for Grades of Veal and Calf Carcasses of the US Department of Agriculture.

(2) Shall be in excellent condition (see 6.5) and free from foreign material (see 4.4.1).

(3) The internal temperature at the thickest portion of the bone-in veal or calf shall be in the temperature range of 28°F (-2°C) to 40°F (4°C) at all times after initial chilling following slaughter and prior to boning or trimming, as applicable.

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3.2.2 Coating ingredients. Only commercially prepared coating ingredients as permitted in the Meat and Poultry Inspection Regulations of the US Department of Agriculture shall be used. Coating ingredients shall be free from foreign materials, off-odors and off-flavors (see 4.4.2).

3.3 Processing.

3.3.1 Boning and trimming requirements. The veal or calf cuts shall be boned and trimmed as follows:

(1) Meat shall be free from foreign material (see 4.4.4).

(2) Meat shall be in excellent condition (see 6.5).

(3) The following shall be removed and excluded from delivery.

(a) Abdominal tunics from flanks only
(b) Serous membranes (membranous coverings)
(c) Exposed (protruding) ligaments and tendons
(d) Gracilis membranes
(e) Kidney and udders
(f) Major lymph glands, i.e., prefemoral, popliteal, prescapular
(g) Thymus glands
(h) Carotid arteries and jugular veins
(i) Dehydrated surfaces (tissue which shows visual evidence of becoming dry)
(j) Tendinous ends of shanks, clods, knuckles, and bottom rounds (to the point where a cross sectional cut exposes at least 75 percent lean tissue)
(k) Bruises, blood clots, bones and cartilage in excess of established tolerances (see 6.5)
(l) Backstrap (ligamentum nuchae)
(m) Hanging tender
(n) The heavy connective tissue (silver strip) on the surface of the outside round adjacent to the knuckle
(o) Periosteum from knuckles, shoulder clods
(p) Sacrosciatic ligament
(q) Bloody neck meat and trimmings

(4) Heavy connective tissue at the scapular joint shall be trimmed even with the lean.

3.3.1.1 Other trimmings. Trimmings other than those prohibited herein, generated as a result of the boning operation and derived from authorized cuts (see 3.2) of the specific batch being processed may be used provided that they are thoroughly mixed with whole cuts.

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3.3.2 Process times and temperatures.

(1) Internal temperature at the thickest part of the chilled bone-in cuts prior to and at start of boning and trimming. 28°F (-2°C) to 40°F (4°C)

(2) Maximum internal temperature at the thickest part of the chilled boneless veal or calf cuts after trimming until the finished product is in the freezing process. 42°F (6°C)

(3) Maximum time after completion of trimming until start of the Comitrol (flaking) process. 72 hours

(4) Maximum time from the start of processing until the formed and frozen cutlets are packaged and packed. 10 days

(5) Maximum temperature of batter in machine during breading (coating) operation. 50°F (10°C)

(6) Internal temperature at the thickest part of the boneless veal or calf cuts and trimmings at the time of flaking. 28°F (-2°C) to 40°F (4°C)

3.3.3 Process requirements for the fabrication of boneless cuts and trimmings.

3.3.3.1 Flake-cut process. Veal or calf cuts and trimmings for the breaded cutlets which comply with the trimming requirements cited in 3.3.1 shall be flaked by impelling at 3600 RPM's, once through an Urschel Comitrol Comminutor, equipped with a size 3K030750D head. The flaked product so processed shall be mechanically mixed to facilitate thorough mixing (see 6.5) and continued until a visual exudate appears on the surface of the meat and the mixture becomes tacky. NOTE: The temperature of the flaked meat shall not exceed 42°F (6°C) during the mixing operation. The use of dry ice is authorized for the purpose of maintaining the required temperature.

3.3.3.2 Other processes. Manufacturing processes and procedures which differ from those specified herein may be used provided the product manufactured therefrom is equivalent in composition and texture to that manufactured under the processes and procedure outlined in paragraph 3.3.3.1 (see 6.4).

3.3.3.3 Additional process requirements.

(1) The flake-cut and mixed product shall be placed in a bag made from 0.0015 inch (0.0381 mm) thick polyethylene film, to prevent drying and freezer burn. The product so packaged shall be frozen to an internal temperature of 0°F (-18°C) or below within 72 hours from the time the product enters the freezer. The product shall be tempered to an internal temperature not to exceed 28°F (-2°C) at the time of pressing (forming).

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In no case during this time shall the product show evidence of soft non-frozen areas on any surface more than 1/4 inch (6 mm) in depth. At no time shall the product be frozen and tempered more than once and the tempered product shall not be trimmed of softened (thawed) areas to comply with the 1/4 inch (6 mm) tolerance for this condition.

(2) The tempered product shall be pressed (formed) in a hydraulic meat press (patty making equipment shall not be acceptable) utilizing a minimum pressure of 500 psi (3447 kPa), to provide cutlets shaped as indicated in figures 1 and 2.

(3) The formed product shall be sliced in equipment designed to provide uniform cutlet weight and thickness as indicated by the finished product requirements. Not more than 1 hour shall elapse from the time the tempered product is pressed until it is sliced.

(4) The cutlets shall be coated in accordance with 3.3.3.4 and packed and returned to the freezer within 1 hour after being sliced.

NOTE: Requirement for Continuous Processing

1. The Comitrol (flaking), mixing, bagging and placement in the freezer shall be in sequence so that as one step is completed, the next step begins in a continuous manner.

2. The tempering, pressing (forming), slicing, packaging, packing and placement in the freezer shall be in sequence so that as one step is completed, the next step begins in a continuous manner.

3.3.3.4 Coating. All breading and batter shall be applied by machine. Hand breading shall not be permitted. Any unused batter or breading in the machine at the end of a day's run shall be discarded. However, used breading may be sifted, when necessary, to clean containers for use the same day. The processing of the cutlets including the coating operation shall be performed in substantially a continuous manner so that the time and temperature limitations specified in 3.3.2 and 3.3.3 are met.

3.4 Finished product. The finished product shall comply with the following requirements.

3.4.1 Breaded cutlets.

(1) Coating shall be free from foreign material (see 6.5) and foreign odor (e.g., musty, moldy, chemical).

(2) Breading shall have a typical fresh breading odor.

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(3) Breading shall have a typical commercially prepared, natural breading color.

(4) Shall be coated with not more than 30 percent coating by weight (see 4.5.1).

(5) Shall weigh 5-1/2 ounces (156 g) plus or minus 1/2 ounce (14 g) and shall be shaped as shown in figure 1.

(6) Shall weigh 3-1/2 ounces (99 g) plus or minus 1/2 ounce (14 g) and shall be shaped as shown in figure 2.

(7) Cutlets shall not be fractured (separated into two or more separate pieces).

(8) There shall be not more than two coating gaps (areas devoid of breading) greater than 1/4 inch (6 mm) in greatest dimension and no coating gap shall be greater than 1/2 inch (13 mm) in greatest dimension.

(9) There shall be no coating ridges (excess breading which projects at the edge of a frozen cutlet longer than 1/2 inch (13 mm).

(10) There shall be no lumps (outcropping of breading on the surface of a frozen cutlet greater than 1/4 inch (6 mm) in depth (thickness).

3.4.2 Unbreaded (decoated) cutlets.

(1) Shall be free from foreign material (see 6.5).

(2) Shall be in excellent condition (see 6.5).

(3) Shall show no evidence of thawing, refreezing or freezer burn and dehydration.

(4) Five and one-half ounce (156 g) cutlets (see figure 1) shall be not less than 3/8 inch (10 mm) nor more than 5/8 inch (16 mm) in thickness.

(5) Three and one-half ounce (99 g) cutlets (see figure 2) shall be not less than 1/4 inch (6 mm) nor more than 1/2 inch (13 mm) in thickness.

(6) Shall be free from bone and cartilage particles in excess of 1/4 inch (6 mm) in any dimension (see 6.5).

(7) Shall have an analytical fat content not greater than 20 percent nor less than 10 percent based on a composite sample.

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3.5 Frozen storage. After the breaded cutlets have been packed and returned to the freezer, the following frozen storage conditions shall be met:

(1) The product shall be held at a uniform temperature of 0°F (-18°C) or colder. The freezer temperature may fluctuate (not to exceed 6°F (-14°C)) for short periods to allow for entering the freezer to restock, withdraw, etc.

(2) The product shall be in excellent condition (see 6.5) and the internal product temperature shall not exceed 0°F (-18°C) at time of shipment and delivery.

(3) The product shall show no evidence of thawing or refreezing at any time.

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.7 Plant qualification. The product shall originate, be processed, marked and handled in plants which are operated under Meat and Poultry Inspection Regulations of the US Department of Agriculture. Finished product refrigeration facilities (freezer, storage and shipping) not operating under these regulations shall meet the requirements of MIL-STD-668.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

(1) First article inspection (see 4.1.2).
(2) Quality conformance inspection (see 4.3).

4.1.2 First article inspection. First article samples of product which the contractor intends to supply shall be inspected to assure compliance with 3.4 and composition and texture requirements (see 3.3.3.2 and 6.3).

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4.2 Plant qualification conditions. The product furnished under this document shall be unacceptable if not produced and stored in plants which currently meet the qualification conditions of 3.7.

4.3 Quality conformance inspection. Inspections required and determination of acceptability shall be in accordance with provisions set forth in MIL-STD-105 except where otherwise indicated hereinafter.

4.4 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.1 Lot assembly by contractor. Prior to Government verification inspection, the contractor shall assemble all meat components of the product (bone-in or boneless) into separate lots. Each lot shall be identified in a manner approved by the supervisory Government inspector. The contractor shall prepare, sign and furnish the Government Quality Assurance Representative (QAR) a report listing the total weight, grade and condition of the lot. Failure to assemble, identify, maintain lot identity and furnish the signed report or failure of any unit (veal or calf component) to comply with 3.2 shall be cause for rejection of the lot or lots, as applicable. With reference to grade, compliance with specified grade requirements shall be deemed as having been met when, upon examination, each unit (veal or calf component) complies with and bears the official United States Department of Agriculture (USDA) grade legend for the grade specified or is accompanied by a grade certificate, as applicable. Nonconformance to any of the above requirements (see 3.2) shall be cause for rejection of the component lot. When foreign materials are found during the examination, the lot shall be referred to the US Department of Agriculture Inspector for disposition in compliance with inspection criteria of the Meat and Poultry Inspection Regulations of the US Department of Agriculture.

4.4.2 Examination of coating ingredient. Examination shall be made to determine compliance with the requirements of 3.2.2. Determination of compliance to the identity requirements shall be ascertained by examination of labels, invoices or other valid documents. The sample to determine compliance with the condition requirements shall be a 2 pound (0.91 kg) composite derived from five primary containers or all the containers if less than five form a lot. Nonconformance to the identity or condition requirements shall be cause for rejection of the component lot or the finished product made therefrom, as applicable. When foreign materials are found during the examination, the lot shall be referred to the US Department of Agriculture Inspector for disposition in compliance with inspection criteria of the Meat and Poultry Inspection Regulations of the US Department of Agriculture.

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4.4.3 Testing of components. Testing of components shall be as shown in table I. Results shall apply to the lot average. Any nonconformance to test requirements shall be cause for rejection of the lot.

TABLE I. Testing of components (see 4.4) 1/

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported to nearest	Test ref and rqmt para
Polyethylene	2 sq yards (1.67 m^2) or equivalent in wraps	Yards (m) or wraps	S-1	Thickness	0.0001 inch (0.001 mm)	3.3.3.3 4.5.3 5.1.1

1/ In lieu of testing for thickness of polyethylene film, determination of compliance to requirements may be ascertained by examination of labels, invoices or other valid documents. Nonconformance to test requirements shall be cause for rejection of the lot.

4.4.4 In-process examination. Examination shall be made during processing to determine compliance with the requirements of 3.3. Records of time and temperature shall be maintained. Nonconformance to one or more of the above referenced requirements reflected by actual examination or by records where applicable shall be cause for rejection of the component lot or the finished product made therefrom. When foreign material is found during the examination, the lot will be referred to the US Department of Agriculture Inspector for disposition in compliance with inspection criteria of the Meat and Poultry Inspection Regulations of the US Department of Agriculture.

4.4.5 Segregation of cuts and trimmings. Examination shall be performed to determine if cuts and trimmings are properly segregated and identified and that such cuts and trimmings are derived from previously accepted lot or lots of component product. Cuts and trimmings shall be identified and segregated in a manner approved by the supervisory inspector. Inadequate identification or segregation or introduction of unaccepted, rejected or prohibited cuts or trimmings shall be cause for rejection of the lot or the resultant end item, as applicable.

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4.4.6 Examination of end item. Classification of defects found during examination shall be in accordance with tables III through VI and 4.4.7 through 4.4.12. Examination criteria for the above referenced tables shall be as shown in table II. The AQLs shall be expressed as defects per hundred units (DHU's) for tables III and V and as percent defective for tables IV and VI. For table IV and V examinations, the cutlets shall first be examined in the breaded state for applicable breading or coating defects. The breading shall then be removed (see 4.5.1.2) and the decoated cutlets examined for applicable defects.

TABLE II. End item examination criteria

Table	Sample unit	Lot size expressed in	Inspection level	AQLs		
				Major A	Major B	Minor
III	One container	Container	S-3	-	4.0	10.0
IV	One breaded cutlet	Pounds (kg)	S-4	-	2.5	6.5
V	One breaded cutlet	Pounds (kg)	S-4	1/	2.5	6.5
VI	One breaded cutlet	Pounds (kg)	S-4	-	-	6.5

1/ Finding of any Major A defect shall be cause for rejection of the lot.

TABLE III. Examination of packing

Category		Defect
Major B	Minor	
151		Liner missing
	201	Polyethylene bag insufficient in size
	202	Bag not folded as specified (see 5.1.1)
	203	Arrangement not as specified (see 5.1.1)

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TABLE III. Examination of packing (cont'd)

Category		Defect
Major B	Minor	
152		Separator missing
153		Tear or hole
154		Separators not placed as specified (5.1.1)
	204	Layer thicker than one cutlet

TABLE IV. Examination of net weight and dimension

Category		Defect 4/
Major B	Minor	5-1/2 oz (156 g) cutlet
151		Net weight less than 5 ounces (142 g) 1/
	201	Net weight more than 6 ounces (170 g) 1/
	202	Thickness less than 3/8 inch (10 mm) 2/ 3/
	203	Thickness more than 5/8 inch (16 mm) 2/ 3/
	204	Shape does not approximate that shown in figure 1
		3-1/2 oz (99 g) cutlet
152		Net weight less than 3 ounces (85 g) 1/
	205	Net weight more than 4 ounces (113 g) 1/
	206	Thickness less than 1/4 inch (6 mm) 2/ 3/
	207	Thickness more than 1/2 inch (13 mm) 2/ 3/
	208	Shape does not approximate that shown in figure 2
		Other weight cutlet (see 6.1)
153		Net weight less than specified
	209	Net weight more than specified
	210	Shape other than specified

1/ Weigh to a tolerance of 1/8 ounce (4 g).

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2/ Measure to a tolerance of 1/16 inch (2 mm).

3/ Examine in the decoated state (see 4.5.1.2).

4/ Examine in the frozen state.

TABLE V. Examination of product characteristics

Category			Defect
Major A	Major B	Minor	
			Breaded cutlets 1/
101			Presence of foreign material 6/
102			Coating has a foreign odor (e.g., musty, moldy, chemical)
	151		Breading lacks a typical fresh odor (e.g., stale odor)
	152		Breading lacks a typical commercially prepared natural color
	153		Cutlet separated into two or more separate pieces
		201	Presence of more than two coating gaps (area devoid of breading) greater than 1/4 inch (6 mm) in greatest dimension 4/
		202	Presence of coating gap (area devoid of breading) in excess of 1/2 inch (13 mm) in greatest dimension 3/
		203	Presence of coating ridges (excess breading which projects at the edge of the frozen cutlets longer than 1/2 inch (13 mm)) 3/
		204	Presence of lumps (outcropping of breading on the surface of a cutlet greater than 1/4 inch (6 mm) in depth (thickness) 5/
			Unbreaded (decoated) cutlets 2/
103			Presence of foreign material 6/
104			Not in excellent condition (see 6.5)
	154		Evidence of thawing, refreezing, freezer burn or dehydration
	155		Presence of bone or cartilage particles in excess of 1/4 inch (6 mm) in any dimension (6.5) 4/

1/ Examine in a frozen state.

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2/ Examine in a thawed state

3/ Measure to a tolerance of 1/4 inch (6 mm)

4/ Measure to a tolerance of 1/8 inch (3 mm)

5/ Measure to a tolerance of 1/16 inch (2 mm)

6/ Within the total quantity represented by all sample units of the sample size specified defects equal to or in excess of the foreign material tolerances specified in 6.5 shall be cause for rejection of the lot.

TABLE VI. Examination for coating percentage

Category	Defect
Minor	
201	More than 30 percent of coating by weight 1/

1/ Record to the nearest whole percent (see 4.5.1)

4.4.7 Examination of shipping container. Examination of shipping container for level B shall be in accordance with the appendix to PPP-B-636 or PPP-B-1163, as applicable. In addition, the following defects shall be included in the examination. Major: (1) Not snug-fitting or evidence of bulging. (2) NSN, contract no., item description or date of pack markings; missing, incorrect or illegible. (3) Color of reinforcing bands not as specified. Minor: (1) Not closed as specified. (2) Other marking requirements not as specified. For level C pack examination shall be in accordance with the aforementioned specifications except only the additional defects as applicable, and closure requirements (see 5.2.2) shall apply.

4.4.8 Segregation and identification of packed production lots. Inspection shall be performed to determine if packed and strapped production lots are kept separated and are distinctly identified. A production lot shall consist of only one weight of finished product (e.g., 5-1/2 oz (156 g) or 3-1/2 oz (99 g), etc.) produced on a given day, shift, etc. Failure to keep production lots separate and distinctly identified until provisionally or finally accepted as applicable, shall be cause for rejection of the involved lot or lots.

4.4.9 Examination at time of freezing and during frozen storage. Examination shall be made at time of freezing and during frozen storage to determine compliance with the applicable requirements of 3.5 as concerns time, temperature

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and fluctuation. Records of time and temperature shall be maintained. Non-conformance to one or more of the above referenced requirements reflected by actual examination or by records shall be cause for rejection of the lot. In addition, evidence of thawing or refreezing shall also be cause for rejection of the lot.

4.4.10 Examination at time of shipment and delivery. Examination shall be made at time of shipment and delivery to determine compliance with the applicable requirements of 3.5. Lot size shall be expressed in terms of the sample unit and the sample unit shall be one filled and closed shipping container. Sample size shall be the number of sample units indicated by inspection level S-2. Temperatures determined to be in excess of 0°F (-18°C) shall be classified as a defective unit and shall be cause for rejection of the lot. In addition, evidence of thawing or refreezing or of product not in excellent condition shall be cause for rejection of the lot.

4.4.11 Examination of shipping container for net weight. The sample unit shall be one filled shipping container. Lot size shall be expressed in terms of the sample unit. Sample size shall be the number of sample units indicated by inspection level S-4. The lot shall be rejected if by using recognized statistical methods, the sample average net weight is determined to be less than the sample average marked net weight.

4.4.12 Sampling procedure and acceptance criteria for testing of cutlets for analytical fat content. Prior to breading, the cutlets shall be tested for fat content as specified in 3.4. Results shall apply to the lot average based on a composite sample. Procedures for testing shall be in accordance with 4.5. The lot size shall represent one day's production. The sample unit for testing shall be a composite of five cutlets, randomly taken and representative of the entire day's production. Test results shall be recorded to the nearest 0.1 percent. Two separate analyses shall be made of the composited sample and result reported to the nearest whole percent as the mean of two readings. Test results less than 10 percent or more than 20 percent shall be cause for rejection of the lot.

4.5 Tests.

4.5.1 Coating content. The coating content for cutlets shall be determined as follows:

4.5.1.1 Skewer method. Remove the unbreaded cutlet from the head of the breading belt, weigh and record the weight. Then pass a skewer (thread or other means of identifying the cutlet) through the end of the cutlet and return the cutlet to the belt just before the first batter dip. Allow the

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cutlet to remain on the belt leading to the packers. Remove the cutlet from the belt at the packers station. Remove the skewer and weigh the breaded cutlet. Calculate the percent coating by the following formula:

$$\text{Percent coating} = 100 - \frac{\text{wt. unbreaded cutlet}}{\text{wt. breaded cutlet}} \times 100$$

4.5.1.2 Washing method.

(1) Equipment

a. Water bath or tap water with aerated faucet spray.
b. Balance accurate to 0.01 ounce (0.28 g).
c. Clip tongs of wire, plastic, or glass.
d. Stop watch or regular watch with second hand.
e. Spatula, 4 inch (102 mm) blade with round tip.
f. Nut picker.
g. US standard sieve - 1/2 inch (13 mm) sieve opening; 12 inch (305 mm) diameter.
h. US standard sieve - ASTM No. 20, 12 inch (305 mm) diameter.
i. Thermometer (immersion type) accurate to ± 2°F (1°C).

(2) Procedure

a. Weigh frozen breaded cutlets, record weight of each.
b. Place each cutlet in a water bath maintained at 63°F (17°C) to 86°F (30°C) (room temperature) and allow to remain until such time as the coating becomes soft and can be easily removed from the still frozen cutlet. Remove coating material by lightly rubbing coated cutlet with fingers, using spatula and nut picker, as applicable, being careful not to damage the cutlet. Place the decoated cutlet on the sieves, 1/2 inch (13 mm) mesh over the No. 20 sieve and pour contents of the container onto them. Remove top sieve and drain at a 45 degree angle for 2 minutes, then transfer to balance. Rinse content of the No. 20 sieve and collect any particles of the cutlet (veal or calf flesh) and add to sample unit on balance and weigh. Alternatively, the above method shall be used utilizing tap water, adjusted to a flow rate of approximately 3 quarts (0.003 m^3) per minute through an aerated faucet spray head. The removal of the coating shall be done over the stacked 1/2 inch (13 mm) mesh and No. 20 sieves.

c. Calculate the percent coating by the following formula:

$$\frac{A-B}{A} \times 100 = \text{percent coating}$$

Code: A = weight of breaded cutlet
B = weight of veal or calf flesh collected on sieves

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4.5.2 Fat analysis. Chemical analysis for fat shall be made in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists cited below:

Fat	Chapter:	Meat and Meat Products	Crude fat using petroleum ether as solvent
	Section:	Meat	

4.5.3 Thickness of plastic film. Thickness of plastic film shall be in accordance with L-P-378.

5. PACKAGING

5.1 Preservation-packaging. The product shall be preservation-packaged in accordance with level B or C, as specified (see 6.1).

5.1.1 Level B. The product shall be layer packed directly into a snug-fitting container (see 5.2 and 6.5) in a pattern and arrangement suited for the weight, shape, and thickness specified (see 1.2, 3.4, 6.1 (for other shapes and weights), and figures 1 and 2). The shipping container shall be lined with a plastic bag constructed from 0.0015 inch (0.0381 mm) polyethylene. The bag shall be of sufficient size to line the interior surfaces of the box completely and shall be constructed so that the opposing lips of the bag have an overlap of not less than 5 inches (127 mm) along the center of the longer axis after the bag is folded down over the packed product. The product shall be arranged in layers with no layer being thicker than one cutlet. Each layer shall be separated from adjacent layers with a single sheet of 0.015 inch (0.381 mm) thick polyethylene not less than 1 inch (25 mm) longer than the width and length of the container. One such sheet shall be placed between the bottom of the bag and the first layer, and another sheet shall be placed on top of the last layer of product.

5.1.2 Level C. The product shall be packaged in accordance with 5.1.1, except that physical requirements for the packaging materials need not apply.

5.2 Packing. Not more than 55 pounds (24.95 kg) of the product shall be packed in a shipping container in accordance with level B or C, as specified (see 6.1). The container shall be packed to maximum capacity without evidence of bulging.

5.2.1 Level B packing. The shipping container shall be a snug-fitting (see 6.5) fiberboard box constructed in accordance with style FTC, V3c, V3s, or V4s, of PPP-B-636. Alternatively, the shipping container shall be a wax-impregnated fiberboard box constructed in accordance with style RSC, class I, type SWCFI, grade 275, of PPP-B-1163. Closure of the wax impregnated box shall

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be with hot melt adhesive in accordance with the appendix of PPP-B-636. All shipping containers shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636. Reinforcing bands shall be uncolored or of a color which contrasts with black. Wax impregnated boxes shall be reinforced with two lengthwise bands and one girthwise band in accordance with the appendix of PPP-B-636.

5.2.1.1 Alternate shipping container. When specified (see 6.1 and 6.7), the shipping container shall be a snug-fitting fiberboard box constructed in accordance with style FTC, W5c or W5s of PPP-B-636. Alternatively, the shipping container shall be a wax-impregnated fiberboard box constructed in accordance with style RSC, class I, type SWCFI, grade 200 of PPP-B-1163. Closure of the wax impregnated box shall be with hot melt adhesive in accordance with the appendix of PPP-B-636. All shipping containers shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636. Reinforcing bands shall be uncolored or of a color which contrasts with black. Wax impregnated boxes shall be reinforced with two lengthwise bands and one girthwise band in accordance with the appendix of PPP-B-636.

5.2.2 Level C packing. The shipping container shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable, except closure of fiberboard boxes shall be in accordance with the appendix to PPP-B-636.

5.3 Marking. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired option offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required (see 5.1 and 5.2).
(c) When packing specified in 5.2.1.1 is required.
(d) Weight and shape of veal cutlets (see 1.2 and figures 1 and 2).
(e) Other weights or shapes desired.

6.2 Plant sanitary conditions. Award of contract for the product specified in this document will be limited to plants known to maintain the required sanitation conditions of 3.7 (also see 4.2).

6.3 First article. When a first article is required for inspection and approval (see 3.1, 4.1.2 and 6.4).

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6.4 First article. The item covered by this specification requires first article test and approval under the appropriate provisions of paragraph 7-104.55 of the Armed Services Procurement Regulations. The first article shall be a preproduction sample. The sample shall consist of 25 cutlets which the contractor shall submit to the US Army Natick Research and Development Command, ATTN: Food Engineering Laboratory, Natick, MA 01760 for evaluation for acceptance. The contracting officer shall include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.5 Definition of terms.

6.5.1 Bruise. A bruise is a damaged portion of the muscle or fatty tissue as evidenced by opaqueness of connective tissue or darkened flesh which exceeds 1 inch (25 mm) in any dimension.

6.5.2 Bone. Bone is connective tissue whose matrix is hardened. Functional definition - bone is ossified material greater than 1/4 inch (6 mm) when measured along the greatest length. Ossified material 1/4 inch (6 mm) or less in length shall not be classified as bone.

6.5.3 Cartilage. Cartilage is gristle or elastic tissue connecting the joints. Functional definition - cartilage is unossified material greater than 1/4 inch (6 mm) when measured along the greatest length. Unossified cartilogenous material 1/4 inch (6 mm) or less in length shall not be classified as cartilage.

6.5.4 Fractured cutlet. A cutlet separated into two or more separate pieces.

6.5.5 Excellent condition. Meat in excellent condition has a degree of freshness corresponding to that exhibited by bone-in meat which is promptly chilled after slaughter to an internal temperature of 28° to 40°F (-2° to 4°C) and maintained within this range with minimal fluctuation and with proper air circulation for a period not exceeding ten days. Meat packaged by heat-shrink vacummized methods may develop a smothered beef odor and/or a darkening of exposed muscle tissue. Fresh bone-in meat may develop surface dehydration and/or a darkening of exposed muscle tissue. Meat displaying any of the undesirable characteristics mentioned above will retain an excellent condition classification only if it can be processed for use without excessive trimming. Meat which has developed a stickiness, sliminess, or discoloration in conjunction with an off-odor shall not be considered in excellent condition. (This definition has been provided by the Department of the Army, Office of the Surgeon General, Washington, DC.)

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6.5.6 Snug-fitting shipping container. A snug-fitting shipping container is one which has no void space in which extra product can be packed or which is not over-filled to a point where the shipping container cannot be stacked safely on a pallet.

6.5.7 Thorough mixing. The mixing together of component materials (i.e., lean with fat, or lean with lean, etc.) in a manner sufficient to uniformly distribute the component materials in a resultant mixture.

6.5.8 Foreign material. Foreign material is any extraneous matter which does not organically belong where found, which has been introduced from the outside, or which does not naturally occur in the quantity found at the location examined.

a. Particles

(1) One or more individual particles, including dust, paint chips, sawdust, and wood splinters in excess of 1/8 inch (3 mm) in greater dimension;

(2) Smaller particles that in an aggregate cover an area of more than 1/2 inch (13 mm) in diameter;

(3) One or more individual particles which are visually determined to be metal, glass, or hard plastic.

b. Soft extraneous material (string, plastic wrap or tape, paper, cardboard, etc.). One or more individual pieces more than 1/2 inch (13 mm) in greatest dimension.

c. Insects. One or more insects, except that one single gnat or fruit fly is considered insignificant and is acceptable.

d. Hair. Five or more single strands of hair; one cluster of hair (strands too numerous to count) in one area.

e. Stains. One or more grease stains or discoloration that individually or in aggregate covers an area more than that of a circle 1/2 inch (13 mm) in diameter.

f. Ingesta. Any amount.

g. Fecal material. Any positive identified amount.

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The final decision with respect to the interpretation of the sanitary, wholesome or esthetic significance of foreign material defects shall remain with a military veterinary officer. (This definition has been provided by the Department of the Army, Health Services Command, Fort Sam Houstan, TX 78234.)

6.5.9 Blood clot. A blood clot is an organized coagulated, opaque mass of blood on the surface of fat or muscle tissue which exceeds 1/2 inch (13 mm) in any dimension.

6.6 Appropriate level of pack. Based on conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/ NAVSUPINST 4030.28/ AFR 71-6/MCO 4030.14D/DSAR 4145.7.

6.7 Intended use of alternate shipping container. Packing specified in 5.2.1.1 is intended for use when experience or knowledge of the destination and mode of shipment indicates that a lower grade of board may be used.

6.8 Supersession data. This specification includes the requirements of Limited Purchase Description LP/P DES 2-74, dated 16 January 1974.

6.9 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

Project No. 8905-0868

Review activities:

Army - MD
Navy - MC, MS
DP - SS

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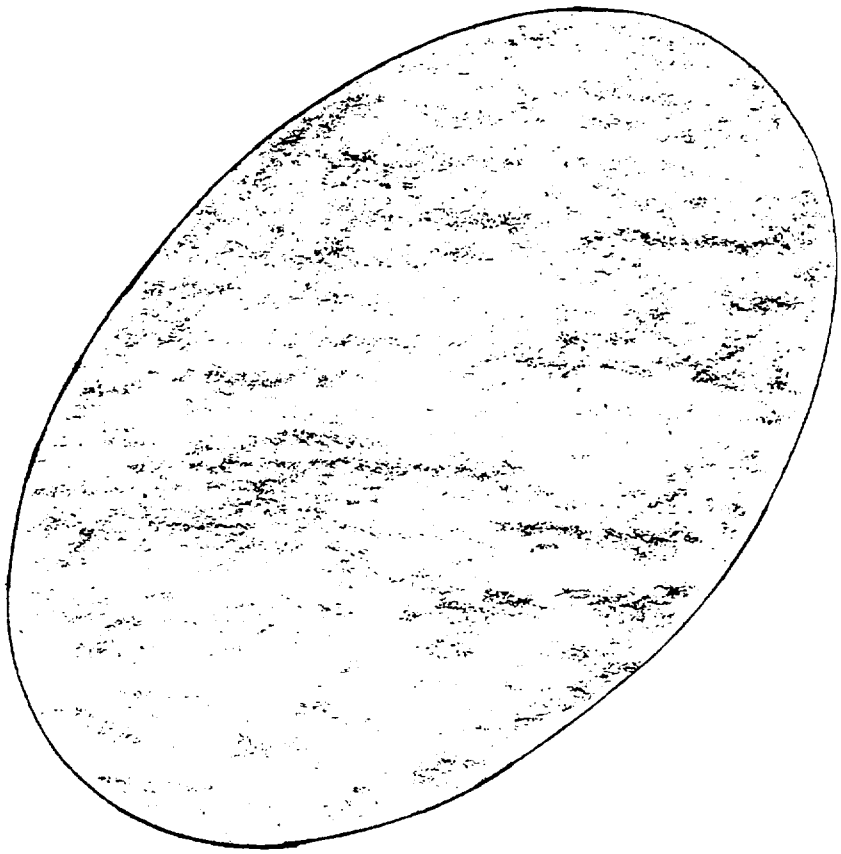


FIGURE 1 VEAL CUTLETS
5 1/2 OZ. (1568)

13-1-164

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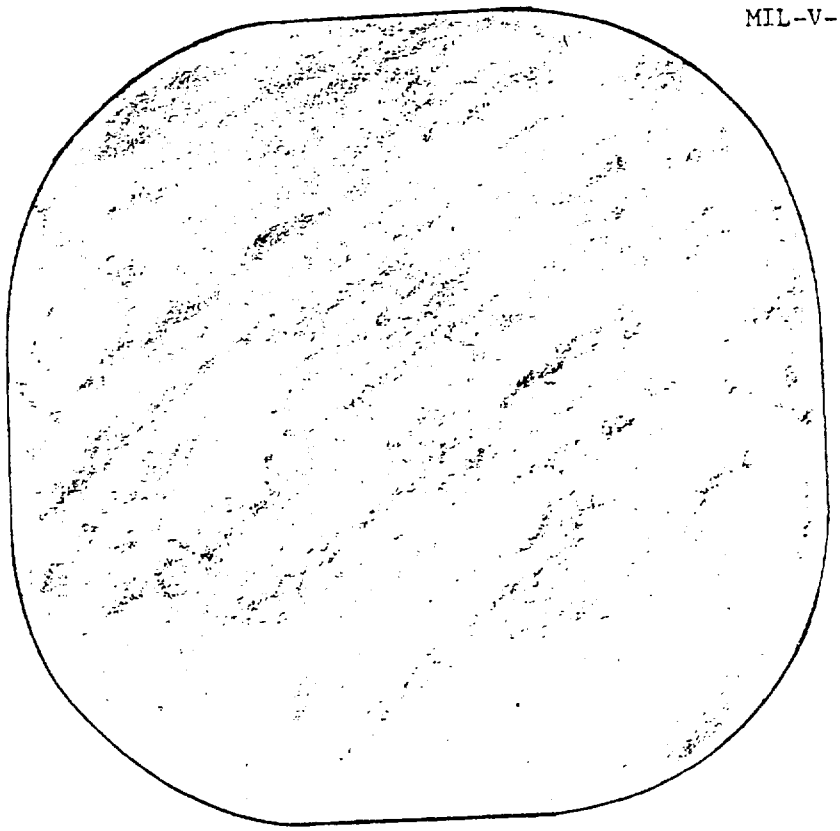


FIGURE 2 VEAL CUTLETS
3 1/2 OZ. (99 g)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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